

Password Strength Analyzer with Custom Wordlist Generator

Internship Project Report

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1. Introduction

Passwords are a fundamental security measure used in protecting digital assets and personal data. However, weak or easily guessable passwords remain one of the biggest causes of data breaches. Attackers often exploit personal information such as names, birthdays, or common patterns to launch brute force or dictionary attacks. This project focuses on analyzing password strength and generating attack-specific wordlists to demonstrate how such weaknesses can be exploited and how stronger passwords can improve security.

2. Objectives

The primary objectives of this project are:

- To analyze the strength of passwords using the zxcvbn library, providing users with scores, guesses, crack time, and feedback.
- To demonstrate how attackers can create custom wordlists based on personal details, increasing the effectiveness of attacks.
- To build user-friendly tools accessible through both command-line interface (CLI) and graphical user interface (GUI).

3. Tools and Technologies Used

The project was developed using the following tools and libraries:

- Python 3.x – The primary programming language used for implementation.
- argparse – For handling command-line arguments in the CLI version.
- zxcvbn – A modern password strength estimator that provides detailed insights and feedback.
- NLTK – Used for word-related processing and handling linguistic data.
- Tkinter – The library used to create an interactive GUI interface.

4. Methodology

The project consists of two main components: a command-line interface tool and a graphical user interface tool. In the CLI tool, users can enter a password, and the system evaluates its strength using zxcvbn. The score, number of guesses, crack time, and feedback are displayed to the user. Users can also input personal details such as names, birth years, or pet names to generate a custom wordlist with variations like leetspeak substitutions, appended numbers, and common patterns. In the GUI version, a user-friendly interface was built using Tkinter, allowing users to enter passwords and receive real-time feedback. Similarly, wordlists can be generated through input fields and saved as text files via file dialogs. This interface ensures accessibility even for non-technical users.

5. Results and Observations

The project successfully demonstrates how password strength can be evaluated and how personal details can be used to create effective wordlists. Sample passwords such as 'Hello@123' and 'Tanmay@123' were analyzed, and the system provided scores and feedback explaining potential weaknesses. The wordlist generation tool created lists containing variations like 'Tanmay2024', 't1ger', and '@Tiger', illustrating how attackers exploit predictable patterns. The GUI tool made it simple for users to interact with the system, improving user experience and understanding.

6. Conclusion

This project provided valuable insights into password security and how attackers exploit weaknesses. By analyzing password strength and generating attack-specific wordlists, users can better understand how to create stronger passwords. The tools developed are practical, user-friendly, and demonstrate important cybersecurity principles. This experience enhanced my understanding of Python programming, libraries like zxcvbn and NLTK, and GUI design using Tkinter.

The tools and methodologies implemented in this project serve as a foundation for further learning and development in cybersecurity practices.